

Problem Statement

Problem Statement: AI-Powered AWS Landing Zone Recommendation System (AWS Control Tower Greenfield Use Case)

You are tasked with building an AI-powered recommendation and guidance system for setting up a secure AWS landing zone in a greenfield environment. The system should intelligently guide cloud architects and platform teams step by step through establishing a multi-account AWS environment using AWS Control Tower, while incorporating security, scalability, governance, and operational guardrails based on AWS best practices.

aws.amazon

+2

Objective

Design and implement an application that:

docs.aws.amazon

+1

Takes organizational and technical inputs such as business units, environments, compliance needs, networking model, identity requirements, and workload isolation needs

aws.amazon

+1

Uses AWS best practices and reference guidance for AWS Control Tower landing zones and multi-account strategies

aws.amazon

+1

Leverages an LLM to generate personalized, human-like recommendations for setting up the landing zone

aws.amazon

+1

Provides step-by-step implementation guidance for account structure, OU design, guardrails, shared services, logging, security, and access management

aws.amazon

+2

Displays clear and actionable recommendations to the user in an easy-to-follow format

docs.aws.amazon

+1

System Workflow

Data Ingestion

Load and preprocess AWS guidance and reference inputs related to landing zones, AWS Control Tower, account design, governance, and guardrails

aws.amazon

+2

Extract relevant knowledge areas such as:

Recommended accounts, for example management, log archive, audit, networking, shared services, and workload accounts

docs.aws.amazon

+1

Organizational unit design such as Sandbox, Infrastructure, and Workloads OUs

aws.amazon

+1

Governance patterns including preventive, detective, and proactive controls in AWS Control Tower

aws.amazon

+1

Identity, logging, and account enrollment considerations

docs.aws.amazon

+1

User Input

Collect user requirements such as:

docs.aws.amazon

+1

Organization type and size

Number of business units or application teams

Environment model, for example dev, test, staging, prod

Compliance or security requirements

Centralized or decentralized networking preference

Identity source and access model

Migration scope, for example greenfield only or future workload onboarding

Additional preferences such as strong guardrails, cost visibility, sandbox isolation, or shared services model

Integration Layer

Interpret and normalize the user inputs into an AWS landing zone design context

aws.amazon

+1

Map the inputs to structured recommendation categories such as OU hierarchy, account vending strategy, networking architecture, logging architecture, identity

federation, and governance controls

aws.amazon

+1

Pass structured findings into an LLM prompt

aws.amazon

Design prompts that help the LLM reason about:

How many accounts should be created

Which OUs should be used

Which controls and guardrails should be applied

How to sequence implementation activities

What dependencies, risks, and validation checks should be considered

aws.amazon

+2

Recommendation Engine

Use the LLM to:

aws.amazon

+2

Recommend a target multi-account structure

Suggest OU hierarchy and account placement

Propose a secure and scalable Control Tower setup plan

Recommend governance guardrails and policy boundaries

Explain why each recommendation fits the user's requirements

Generate a phased implementation plan such as foundation, governance, networking, identity, security, and onboarding

aws.amazon

+1

Output Display

Present recommendations in a user-friendly format such as:

aws.amazon

+1

Recommended OU structure

Core shared accounts to create

Workload account model

Security and logging design

Guardrails and governance recommendations

Identity and access setup guidance

Networking model recommendations

Step-by-step implementation plan

AI-generated explanation for each recommendation

Risks, assumptions, and next actions

Example Deliverable Format

The application output may look like this:

```
aws.amazon  
+2
```

Recommended OUs: Security, Infrastructure, Sandbox, Workloads-NonProd,
Workloads-Prod

Core Accounts: Management, Log Archive, Audit, Networking, Shared Services

Guardrails: Mandatory preventive controls for production OUs, detective controls for logging and compliance, tighter SCP strategy for sandbox and production separation

Implementation Steps: Enable AWS Control Tower, configure IAM Identity Center, define OU hierarchy, create shared accounts, enable centralized logging, apply controls, provision workload accounts, validate compliance

AI Explanation: This structure improves blast-radius isolation, supports governance by OU, separates shared services from workloads, and creates a scalable pattern for future onboarding